



Evidence Analysis

1. Introduction

Digital evidence analysis is a critical component of incident response. After detecting and containing a security incident, investigators must collect, preserve, and analyze digital artifacts to determine the scope of compromise and identify attacker behavior.

Proper evidence handling ensures integrity, admissibility, and forensic reliability. This report demonstrates structured evidence analysis using:

- **Velocidex Enterprises Velociraptor** (Endpoint Forensics)
- **FTK Imager** (Disk and Memory Evidence Collection)

The task involved analyzing suspicious network connections and maintaining a documented chain of custody.

2. Objective

The objectives:

- Collect endpoint forensic data.
 - Identify suspicious network connections.
 - Preserve evidence integrity.
 - Maintain proper chain-of-custody documentation.
 - Analyze artifacts to support investigation findings.
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3. Evidence Collection Process

3.1 Evidence Sources

The following evidence sources were collected from a Windows virtual machine:

- Network connection logs
 - Running process list
 - System event logs
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- Disk image snapshot

Evidence collection was performed in a controlled lab environment to simulate real-world forensic procedures.

3.2 Tool Usage

1. Velociraptor: Used for endpoint artifact collection and querying.

Query executed: `SELECT * FROM netstat`

Purpose:

- Identify active network connections.
- Detect suspicious external communications.
- Review listening ports.

2. FTK Imager

Used to:

- Create forensic disk image.
 - Preserve file system artifacts.
 - Generate hash values for integrity verification.
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4. Evidence Analysis Findings

4.1 Network Analysis Results

The netstat query revealed:

- Multiple established connections.
- One outbound connection to unfamiliar external IP.
- Suspicious process associated with network activity.

Suspicious Observation:

- External IP not recognized within trusted infrastructure.
- Connection initiated by unusual PowerShell process.
- Port used: 4444 (commonly used in reverse shells).

This behavior suggests potential command-and-control (C2) activity.

4.2 Process Analysis

Additional Velociraptor queries revealed:



- PowerShell process executed with encoded command.
- Process initiated shortly after phishing alert.
- Parent process linked to user session.

This correlates with previously detected malicious activity.

5. Chain of Custody Documentation

Maintaining chain of custody ensures evidence integrity and accountability.

Evidence Log

Item	Description	Collected By	Date	Hash Value
Network Log	Server-Z Network Artifact	SOC Analyst	2025-08-18	SHA256-ABC123
Disk Image	Windows VM Snapshot	SOC Analyst	2025-08-18	SHA256-DEF456

5.1 Chain-of-Custody Principles Followed

- Evidence collected in read-only mode.
- Hash values generated immediately after acquisition.
- Documentation completed at time of collection.
- No modification performed on original artifacts.
- Access restricted to authorized personnel.

These steps preserve evidentiary integrity.

6. Integrity Verification

Hash values were calculated using SHA-256 algorithm.

Integrity verification ensures:

- No tampering occurred.
- Evidence remains unchanged.
- Forensic reliability maintained.

Re-hashing after analysis confirmed identical hash values.



7. Analysis Summary

The investigation revealed:

- Suspicious outbound network connection.
- Reverse shell-like communication pattern.
- PowerShell process with encoded execution.
- Potential command-and-control channel.

Although contained during earlier response, forensic artifacts confirm malicious behavior occurred.

8. Evidence Summary

Digital forensic analysis identified suspicious outbound communication initiated by a PowerShell process. Network connections indicated potential reverse shell activity over port 4444. Evidence was collected using Velociraptor and preserved via FTK Imager. Hash-based integrity verification ensured forensic reliability. Findings support earlier detection of malicious activity.

9. Challenges Identified

- Large volume of endpoint data.
 - Need for correlation with earlier alerts.
 - Distinguishing legitimate PowerShell usage from malicious activity.
 - Ensuring minimal disruption during collection.
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10. Recommendations

- Enable enhanced PowerShell logging.
 - Monitor outbound traffic to non-whitelisted IPs.
 - Implement EDR behavioral monitoring.
 - Regularly test forensic readiness.
 - Maintain updated chain-of-custody templates.
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11. Conclusion

This task successfully demonstrated digital evidence collection, analysis, and preservation within a structured SOC workflow. By using forensic tools and maintaining strict chain-of-custody procedures, evidence integrity was preserved while identifying suspicious network behavior. Proper forensic methodology strengthens investigative reliability and enhances SOC incident response maturity.